

Groups of minimal degree three

Alternating blocks, binary codes, and cyclic actions

EULER

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The original problem. The minimal degree of a permutation group is the least number of points moved by a nonidentity element. Montero and Potočník ask for a detailed description of the transitive groups of minimal degree three, including which groups between $\text{Alt}(m)^k$ and $\text{Sym}(m) \wr \text{Sym}(k)$ are transitive [1, Problem 2].

The key is that a three-cycle reveals its own block. Once these blocks are recognized, only the signs of the permutations inside them remain to be chosen. We first give a complete description, including when two choices give the same permutation group. We then make that answer explicit when the blocks are permuted cyclically. The latter result holds for every number of blocks; even lengths introduce one genuine extra choice.

1 Recover the blocks before choosing coordinates

Join two points when they occur in the support of a common three-cycle of the group, and take connected components. This construction uses the action itself, so conjugation must carry its components to components.

Lemma 1. *If a transitive group has minimal degree three, these components form equally sized blocks, each of size at least three, and the group contains the full alternating group on each block, acting trivially elsewhere.*

Proof. An element moving three points is a three-cycle. The union of the supports of all such elements is nonempty and invariant, hence contains every point. Conjugation preserves the collection of supports, so transitivity also makes all its connected components equally large.

It remains to explain why a connected collection of three-cycles generates the alternating group on its points. Begin with one support. Suppose the alternating group on a set B has already been generated and another three-cycle meets B . If it brings in one new point, conjugation by $\text{Alt}(B)$ supplies the three-cycles needed to adjoin that point. If it brings in two new points u, v , write it as (xuv) and conjugate it to (yuv) for another $y \in B$. The product of the first cycle and the inverse of the second is (xuy) . We may therefore adjoin u first, and then v . This argument also works when $|B| = 3$: a three-cycle and its inverse supply both orientations of each pair, and conjugation cycles the three old points. Adding intersecting supports along a connected component proves the claim. All the permutations just used are supported on that component alone. $\square$

Number the blocks and their points. If there are k blocks of size m , the original action now lies between $\text{Alt}(m)^k$ and $\text{Sym}(m) \wr \text{Sym}(k)$. This gives the reduction required by the original problem without a classification theorem for primitive groups.

2 The complete description

Record the sign in each block as a vector in $V = \mathbb{F}_2^k$: zero means even and one means odd. After factoring out $\text{Alt}(m)^k$, a permutation is a pair (v, h) consisting of its signs and its permutation of the blocks. The multiplication rule is

$$(v, h)(w, l) = (v + hw, hl).$$

For a subgroup of these pairs, let H be its action on the blocks and let K be the sign vectors occurring above the identity block action. Then K is a binary linear code preserved by H . Above any fixed $h \in H$, the permitted signs form a single coset of K ; denote it by $c(h)$. Closure under multiplication says exactly that

$$c(hl) = c(h) + hc(l) \quad \text{in } V/K. \quad (1)$$

Such a function is usually called a cocycle. Here it simply records which sign coset accompanies each movement of the blocks.

Theorem 2. *Every transitive permutation group of minimal degree three is obtained by choosing a block size $m \geq 3$, a number of blocks $k \geq 1$, a transitive group H on those blocks, a proper H -invariant binary code K , and a function $c : H \rightarrow V/K$ satisfying (1). Take all block permutations $h \in H$ with signs in the coset $c(h)$, and all their lifts to $\text{Sym}(m) \wr \text{Sym}(k)$.*

Conversely, every such choice gives a transitive group of minimal degree three. Two choices give conjugate permutation groups precisely when their block sizes and numbers agree and, after relabelling the blocks, their codes agree and their functions differ by

$$c(h) \mapsto c(h) + b + hb \quad (b \in V/K). \quad (2)$$

The relabelling must carry one block-action group to the other.

Proof. The preceding discussion and Lemma 1 give all the data from a group. Conversely, (1) makes the prescribed pairs a subgroup, and taking its full inverse image supplies $\text{Alt}(m)^k$. This alternating subgroup is transitive inside every block, so the whole group is transitive exactly when H is transitive on the blocks.

A three-cycle inside one block is always present. A permutation moving only two points would be a transposition inside one block, with sign vector one of the coordinate vectors. An invariant code for a transitive H contains one coordinate vector only if it contains them all. Thus such a transposition exists exactly when $K = V$. A proper code gives minimal degree three, as required.

For the equivalence statement, all three-cycles in the constructed group are confined to one block: moving blocks moves at least $2m$ points, and acting nontrivially in two blocks moves at least four. Conversely, every three-cycle inside a block is present. Hence the blocks are exactly the intrinsic components used in Lemma 1. Any conjugator must relabel those blocks, and then permute points within them.

Relabelling by s sends H to sHs^{-1} , K to sK , and the coset above h to the relabelled coset above shs^{-1} . Conjugating inside the blocks by a permutation with sign vector b changes the sign of (v, h) to $v + b + hb$. This is precisely (2). Every vector b can be realized by choosing a transposition in each indicated block. The stated equivalences are therefore both necessary and sufficient. $\square$

This also gives a list without repetitions. Choose H up to relabelling; choose K up to the relabellings preserving H ; then choose c up to (2) and the remaining relabellings preserving both H and K . No simplicity assumption on $\text{Alt}(m)$ is used, so the small block sizes three and four require no exceptions.

3 A cyclic action: the natural boundary is every length

Suppose the block action is a regular cycle of length k . Regard a sign vector as a polynomial modulo $X^k - 1$; shifting the blocks is multiplication by X . A shift-invariant code is then an ideal, uniquely generated by a monic divisor g of $X^k - 1$. The full code has $g = 1$ and must be excluded.

Write $k = ed$, where d is odd and e is a power of two. Over $\mathbb{F}_2$,

$$X^k - 1 = (X^d - 1)^e.$$

Let s be the exponent of $X + 1$ in g , so $0 \leq s \leq e$.

Theorem 3. *For a regular cyclic block action, the complete classification is as follows. Choose a proper cyclic code, up to a multiplier of the block positions. There is one group for that choice unless $0 < s < e$; in this intermediate case there are exactly two, distinguished by a zero or nonzero twist.*

In particular, odd lengths have no twist. At a power-of-two length k there are exactly $2k - 1$ classes, for every block size $m \geq 3$.

Proof. Let t be the cyclic shift and put $M = V/K$, which is the polynomial module $\mathbb{F}_2[X]/(g)$. A cocycle is determined by the single value $a = c(t)$. Going once around the cycle and using (1) gives

$$(1 + X + \cdots + X^{k-1})a = 0.$$

Changing coordinates as in (2) changes a by a multiple of $X + 1$. Thus we need only take the solutions of this displayed equation modulo $(X + 1)M$.

Factor g into powers of irreducibles and use the Chinese remainder theorem. On every factor other than $X + 1$, multiplication by $X + 1$ is invertible, so that factor contributes no choice. On the remaining factor put $Y = X + 1$. Its module is $\mathbb{F}_2[Y]/(Y^e)$, and the sum $1 + X + \cdots + X^{k-1}$ is Y^{e-1} times an invertible element. If $s = 0$, there is nothing to choose. If $s = e$, its kernel is YM , which is exactly what the coordinate change removes. If $0 < s < e$, the sum acts as zero, and quotienting by YM leaves one binary choice. These are all the cases.

A permutation normalizing the regular cyclic action has the form $i \mapsto ai + b$, where a is a unit modulo k . Translations already preserve every cyclic code, leaving just the multipliers. Their action on cocycles is linear and carries zero to zero. When the space of choices is one-dimensional over $\mathbb{F}_2$, its unique nonzero element must therefore remain nonzero. The two twists cannot become conjugate. This proves the classification, including its equivalences.

When k is a power of two, the proper codes have generators $(X + 1)^s$ for $1 \leq s \leq k$. The first $k - 1$ choices each give two groups and the last gives one. Their total is $2k - 1$. $\square$

The twist has a simple meaning at two blocks. For the repetition code $K = \{00, 11\}$, the quotient on signs and blocks is either the Klein four group or the cyclic group of order four: the lifted block swap has square zero or the nonzero repetition vector. The zero code gives a third class. The even case is therefore a real addition to the classification, rather than an exceptional convention in the counting formula.

4 Counting the classes

For completeness the same description gives an explicit finite count at every length. Let $\mathcal{D}_d$ be the doubling orbits on the residues modulo the odd number d . A unit a modulo d permutes these orbits; write $c_d(a)$ for its number of cycles on $\mathcal{D}_d$.

Corollary 4. *The number of classes with a regular cyclic block action of length $k = ed$ is*

$$\frac{1}{|(\mathbb{Z}/d\mathbb{Z})^*|} \sum_{a \in (\mathbb{Z}/d\mathbb{Z})^*} 2e(e+1)^{c_d(a)-1} - 1.$$

At $d = 1$, the unit action has one element and $c_1 = 1$.

Proof. The irreducible factors of $X^d - 1$ correspond to the doubling orbits: Frobenius squares the d th roots of unity. A divisor of $(X^d - 1)^e$ chooses an exponent from 0 through e for each orbit. A multiplier-fixed choice is constant on each of its cycles. The orbit of zero corresponds to $X + 1$ and is fixed separately. Its two endpoint exponents each have one twist choice, and its $e - 1$ interior exponents each have two, giving $2e$ choices in total. Each remaining cycle has $e + 1$ choices. All units modulo d lift to units modulo k , so this is the full multiplier action. Averaging the fixed choices over that action counts its orbits. Finally remove the full code, which was counted once. $\square$

For example, at seven blocks there are three doubling orbits, $\{0\}$, $\{1, 2, 4\}$ and $\{3, 5, 6\}$. Multipliers may interchange the last two. A proper code is specified by whether the first orbit is chosen and how many of the other two are chosen, excluding the empty choice: there are $2 \cdot 3 - 1 = 5$ classes. The counts for $k = 1, \dots, 9$ are 1, 3, 3, 7, 3, 11, 5, 15, 7.

Sources and scope

The original quotient criterion and labelled parametrization are in the released EULER case [2]. Here the original problem and its proof are included before passing to conjugacy and to all cyclic lengths. The elementary cocycle and cyclic-code tools are classical; these applications are not a claim of new general machinery or certified priority. The code equivalence is for a code together with its specified cyclic top action, not unrestricted permutation equivalence of cyclic codes. The classification concerns permutation conjugacy, not abstract group isomorphism.

References

- [1] A. Montero and P. Potočník, *Vertex-transitive graphs with small motion and transitive permutation groups with small minimal degree*, Problem 2, arXiv:2405.10088v2.
- [2] EULER, Minimal-degree-three imprimitive groups.
- [3] N. Aydin and R. O. VandenBerg, *A New Algorithm for Equivalence of Cyclic Codes and Its Applications*, arXiv:2107.00159.